Answer Key - Cybersecurity - Graduate Level Assessment 17 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) Phishing; B) MiTM; C) Buffer-overflow; D) Clickjacking
   Note: Correct option: C - Buffer-overflow

2. **Answer:** A
   Options: A) To identify live systems; B) To locate live systems; C) To identify open ports; D) To locate firewalls
   Note: Correct option: A - To identify live systems

3. **Answer:** C
   Options: A) Active, inactive, standby; B) Open, half-open, closed; C) Open, filtered, unfiltered; D) Active, closed, unused
   Note: Correct option: C - Open, filtered, unfiltered

4. **Answer:** C
   Options: A) C, Ruby; B) Python, Ruby; C) C, C++; D) Tcl, C#
   Note: Correct option: C - C, C++

5. **Answer:** D
   Options: A) Yes, if the PRG is really "secure"; B) No, there are no ciphers with perfect secrecy; C) Yes, every cipher has perfect secrecy; D) No, since the key is shorter than the message
   Note: Correct option: D - No, since the key is shorter than the message

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'buffer'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'Restrict what operations/data the user can access'.

10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'No string boundary checks in predefined functions'.

Long Form Response

11. **Answer:** def occurance_substring(text,pattern): | return (text[s:e], s, e)
Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** def parabola_directrix(a, b, c): | return directrix
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key